Grade 10 Electricity Memo

[60 marks]

1.1 B ✓✓
1.2 A ✓✓
1.3 C ✓✓
1.4 C ✓✓ [8]

2.1 Current is the rate of flow of charge. ✓✓ (2)

2.2.1 $I = \frac{Q}{t} \checkmark = \frac{3,5}{8 \times 60} = \frac{3,5}{480} \checkmark = 7,29 \times 10^{-3} \checkmark A$ (3)

2.2.2 $W = VQ \checkmark$
$= VIt \checkmark$
$= 220 \times 7,29 \times 10^{-3} \times 480$
$= 770 \ J \checkmark$ (3)
[8]

3.1 $R_T = \frac{V_T}{I_T} \checkmark = \frac{9}{4,5} \checkmark = 2\Omega \checkmark$ (3)

OR $R_T = \frac{\text{prod}}{\text{sum}} = \frac{6 \times 3}{6+3} = \frac{18}{9} = 2\Omega$

3.2 $V = 9V$ ✓✓ (2)

3.3 (A1) $I = \frac{V}{R} = \frac{9}{6} \checkmark = 1,5 \checkmark A$

(A2) $I = \frac{V}{R} = \frac{9}{3} \checkmark = 3 \checkmark A$ (4)

OR

	TOP : BOT
R	6 : 3
I	3 : 6

3 V na · $\frac{6}{3} \times 4$ C

3.4 $W = VQ \checkmark$
$= 9 \times 3 \checkmark$
$= 27 \ J \checkmark$ (3)
[12]

4.1 $R_p = \frac{\text{prod}}{\text{sum}} \checkmark = \frac{18 \times 6}{18+6} \checkmark = \frac{108}{24} = 4,5 \checkmark \Omega$ (3)

OR $\frac{1}{R_p} = \frac{1}{18} + \frac{1}{6}$
$= \frac{1+3}{18} = \frac{4}{18}$
$R_p = \frac{18}{4} = 4,5\Omega$

4.2 $R_T = R_p + 3$
$= 4,5 + 3$
$= 7,5\Omega \checkmark$ (1)

4.3 (A) $I = \frac{V}{R} \checkmark = \frac{12}{7,5} \checkmark = 1,6 \checkmark A$ (3)

4.4 (V) $V = IR \checkmark$
$= 1,6 \times 4,5 \checkmark$
$= 7,2 \ V \checkmark$ (3)

OR $V_{3\Omega} = IR = 1,6 \times 3 = 4,8V$

$V_{//} = V_T - V_{3\Omega} = 12 - 4,8 = 7,2V$

[10]

(3)

5.1 $R_p = \frac{\text{prod}}{\text{sum}} = \frac{4 \times 8}{4+8} = \frac{32}{12} = 2,67\ \Omega$

OR $\frac{1}{R_p} = \frac{1}{4} + \frac{1}{8} = \frac{2+1}{8} = \frac{3}{8}$

$R_p = \frac{8}{3} = 2,67\ \Omega$

$V_{//} = IR$
$= 3 \times 2,67$
$= 8,01\ V$

(5)

5.2 $V_R = 18 - 8$
$= 9,99 V$

(2)

5.3 $R = \frac{V}{I} = \frac{10}{3} = 3,33\ \Omega$

(2)

5.4 $W = VIt$
$= 10 \times 3 \times 35$
$= 1050\ J$

(3)
[12]

6.1 Length of wire
thickness of wire

(2)

6.2 $R = \text{grad } A = \frac{y_2 - y_1}{x_2 - x} = \frac{6 - 2,1}{0,54 - 0,2} = \frac{3,9}{0,34} = 11,47\ \Omega$

$B = \frac{3,4 - 1,1}{0,6 - 0,2} = \frac{2,3}{0,4} = 5,75\ \Omega$

(5)

(4)

6.3 'A'

'A' has a higher resistance ∴ more work done for charge to move through resistor. i.e. more electrical energy transferred into heat energy.

$W = VQ$ but $V = IR$
∴ $W = IRQ$ ✓equation

(3)
[10]

OR Can say for

$V = I_c R$
$= \uparrow R \uparrow V \therefore \uparrow W$